

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I P. Subendran Kumar Reddy (192425224) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Subendran

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

2. Given

System 1

→ IP address : 192.168.1.130

→ Subnet mask : 255.255.255.192 (126)

System 2

→ IP address : 192.168.1.190

→ Subnet mask : 255.255.255.192 (126)

Step 1: Understand the subnet mask

The subnet mask 255.255.255.192 is equivalent to 126.

A 126 subnet means

→ Total IP addresses = 64

→ Network address = 1

→ Broadcast address = 1

→ Usable host address = $64 - 2 = 62$

The subnet increment is 64, so the subnet ranges are:

Subnet	Network Address	Broadcast Address
1	192.168.1.0	192.168.1.63
2	192.168.1.64	192.168.1.127
3	192.168.1.128	192.168.1.191
4	192.168.1.192	192.168.1.255

Step 2: Analyze system 1

The IP address of system 1 is 192.168.1.130.

This address falls within the subnet

192.168.1.128 - 192.168.1.191

Therefore,

→ Network address : 192.168.1.128

→ First valid host : 192.168.1.129

→ Last valid host : 192.168.1.190

→ Broadcast address : 192.168.1.191

since 192.168.1.130 lies between 192.168.1.129 and 192.168.1.190, it is valid host IP address.

step 3: Analyze system 2

The IP address of system 2 is 192.168.1.190

It also belongs to the subnet

192.168.1.128 - 192.168.1.191

Therefore

→ network address : 192.168.1.128

→ first valid host : 192.168.1.129

→ last valid host : 192.168.1.190

→ Broadcast Address : 192.168.1.191

since 192.168.1.190 is the last usable host before the broadcast address, it is also called valid host IP address

step 4: Number of usable hosts

formula :

$$\begin{aligned} \text{Usable Hosts} &= 2^{(32-26)} - 2 \\ &= 2^6 - 2 = 64 - 2 = 62 \\ &= 62 \text{ usable hosts} \end{aligned}$$

Final Result

parameter	system 1	system 2
IP address	192.168.1.130	192.168.1.190
subnet mask	/26	/26
Network Address	192.168.1.128	192.168.1.128
valid Host Range	192.168.1.129 - 192.168.1.190	192.168.1.129 - 192.168.1.190
Broadcast Address	192.168.1.191	192.168.1.191
usable Hosts	62	62
valid Host?	Yes	Yes

3.

Given

the organization has been assigned the network
block : 172.16.0.0/16

the required subnet masks are;

→ HR → /25

→ Sales → /26

→ IT → /27

→ Admin → /28

Step 1: HR Department (/25)

A /25 subnet contains:

Total addresses = 128

usable hosts = 126

Allocated subnet:

→ Network address : 172.16.0.0

→ Host Range : 172.16.0.1 - 172.16.0.126

→ Broadcast address : 172.16.0.127

→ usable hosts : 126

Step 2: Sales Department (/26)

The next available subnet begins at 172.16.0.128

A /26 subnet contains:

→ Total addresses = 64

→ usable hosts = 62

Allocated subnet:

→ Network address : 172.16.0.128

→ Host Range : 172.16.0.129 - 172.16.0.190

→ Broadcast address : 172.16.0.191

→ usable hosts : 62

Step 3 ÷ IT department

The next available subnet starts at 172.16.0.192

A /27 subnet contains

→ Total address = 32

→ Usable hosts = 30

Allocated subnet :

→ Network address : 172.16.0.192

→ Host addresses : 172.16.0.193 - 172.16.0.222

→ Broadcast address : 172.16.0.223

→ Usable hosts : 30

Step 4 ÷ Admin Department (/28)

The next available subnet starts at 172.16.0.224

A /28 subnet contains :

→ Total addresses = 16

→ Usable hosts = 14

Allocated subnet :

→ Network address : 172.16.0.224

→ Host range : 172.16.0.225 - 172.16.0.238

→ Broadcast address : 172.16.0.239

→ Usable hosts : 14

Summary Table

	HR	Sales	IT	Admin
Subnet	/25	/26	/27	/28
Network address	172.16.0.0	172.16.0.128	172.16.0.192	172.16.0.224
Host Range	172.16.0.1 - 172.16.0.126	172.16.0.129 - 172.16.0.196	172.16.0.193 - 172.16.0.222	172.16.0.225 - 172.16.0.238
Broadcast Address	172.16.0.127	172.16.0.191	172.16.0.223	172.16.0.239
Usable hosts	126	62	30	14